

Population aging and intergenerational relationships

Economic implications of population aging

- Most of the challenges (and alarmism) about aging are related to the economy.
- Sustainability of Pensions and Social Transfers
- Rising of Health costs
- Who is going to Care?



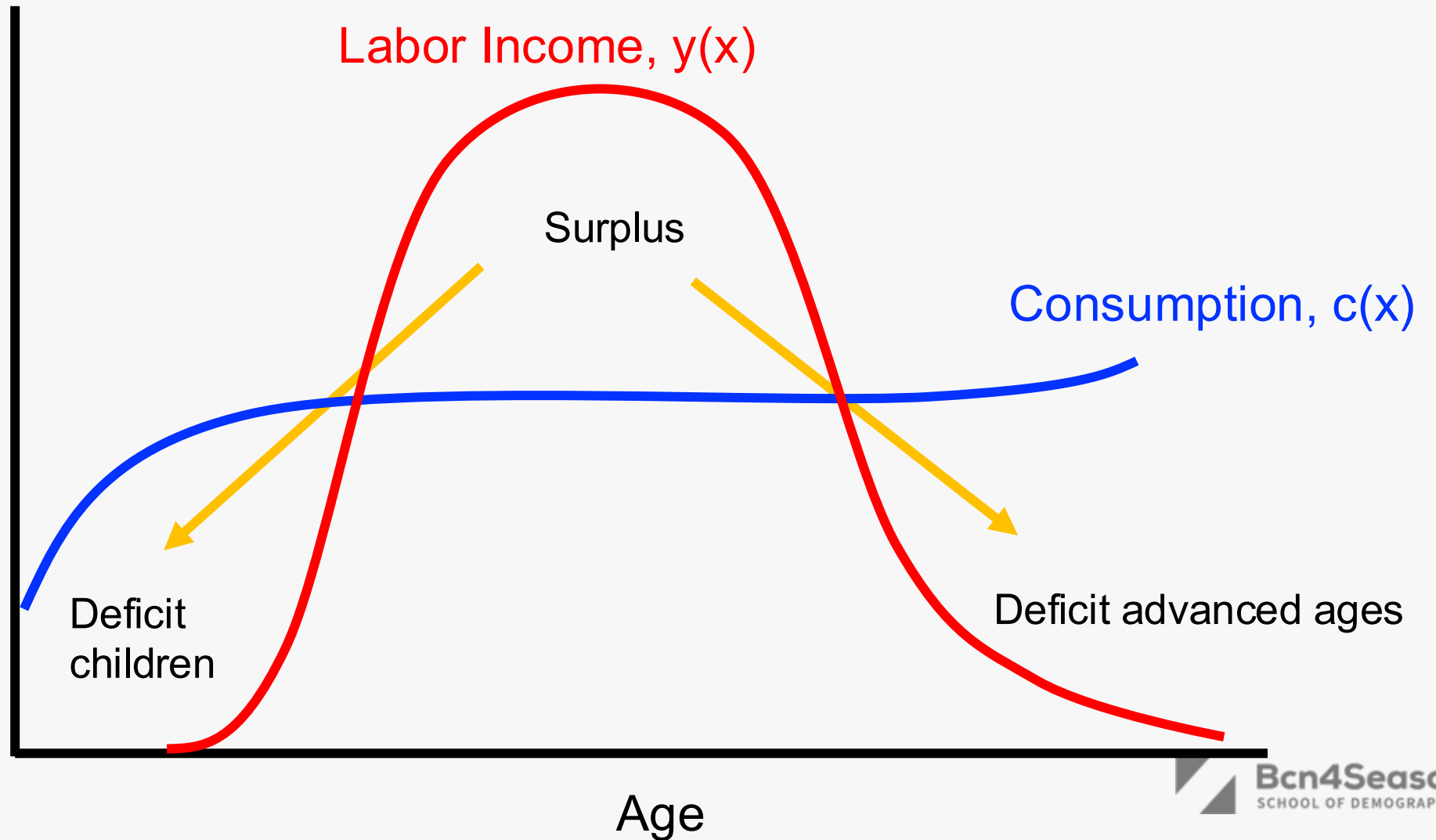
National Transfers Accounts

- Approach from Ronald D. Lee and Andrew Mason
- Methodology to disaggregate National Accounts by age
- This allows to:
 - Understand how resources flow from one age group to the other
 - Assess the economic impact of population aging in a more realistic way



Lifecycle Deficit

Per capita
value



Life Cycle Deficit

Consumption (C) – Labor Income (YL)

↗ PRIVATE Transfers
(TF)

→ PUBLIC Transfers
(TG)

↘ ASSET BASED
Reallocations (YA)



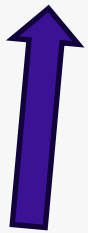
Lifecycle Deficit formula

$$C - YL = (TG^{+} - TG^{-}) + (TF^{+} - TF^{-}) + (YA - S)$$



Lifecycle Deficit formula

$$C - YL = (TG^+ - TG^-) + (TF^+ - TF^-) + (YA - S)$$



Lifecycle deficit



Lifecycle Deficit formula

$$C - YL = (TG^+ - TG^-) + (TF^+ - TF^-) + (YA - S)$$



Public transfers



Lifecycle Deficit formula

$$C - YL = (TG^+ - TG^-) + (TF^+ - TF^-) + (YA - S)$$



Private transfers



Lifecycle Deficit formula

$$C - YL = (TG^+ - TG^-) + (TF^+ - TF^-) + (YA - S)$$



Asset – Based Reallocations



National Accounts equivalent to economic profiles

NTA profile	NA macro control
Consumption: Private and Public	Household Final Consumption Expenditure; Government Final Consumption Expenditure
Labor Income: Earnings and Self-employment income	Compensation of Employees; mixed income adjusted to isolate labor component
Public Transfers: Outflows and Inflows (cash and in kind)	Taxes and social contributions; social benefits; pensions; education and health transfers;
Private Transfers: interhousehold and intrahousehold	Remittances and inferred residually
Asset-Based Reallocations: asset income, savings,	Property income; operating surplus; household saving; consumption of fixed capital; entrepreneurial income



Adjustment to the National Accounts macro-control

$$X = \sum_a x(a) \cdot P(a)$$

$$x_{NA}(a) = x(a) \cdot \frac{X}{X_{NA}}$$



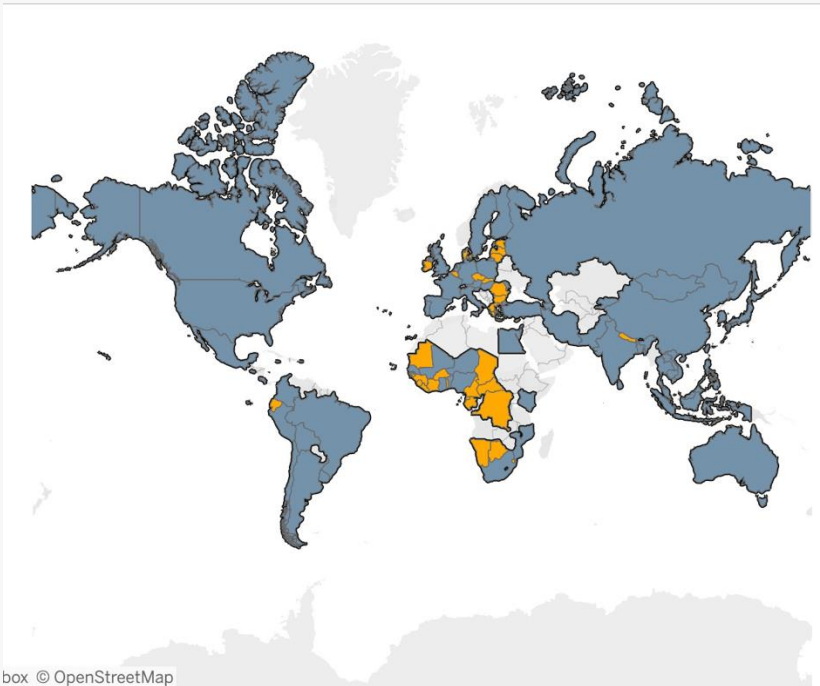
Adjustment to the National Accounts macro-control

Aggregate profile $\rightarrow X = \sum_a x(a) P(a)$ $\leftarrow$ Per capita profile

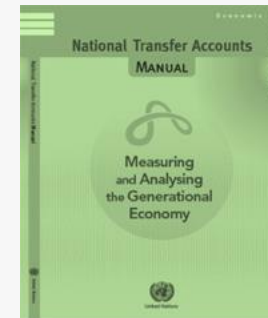
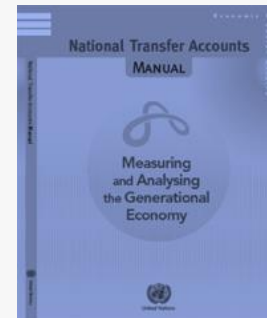
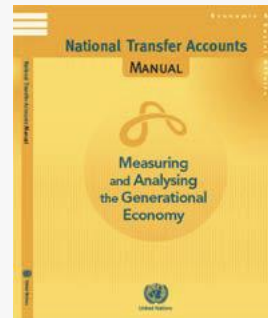
Adjusted per capita profile $\rightarrow x_{NA}(a) = x(a) \cdot \frac{X}{X_{NA}}$ $\leftarrow$ Macro-control



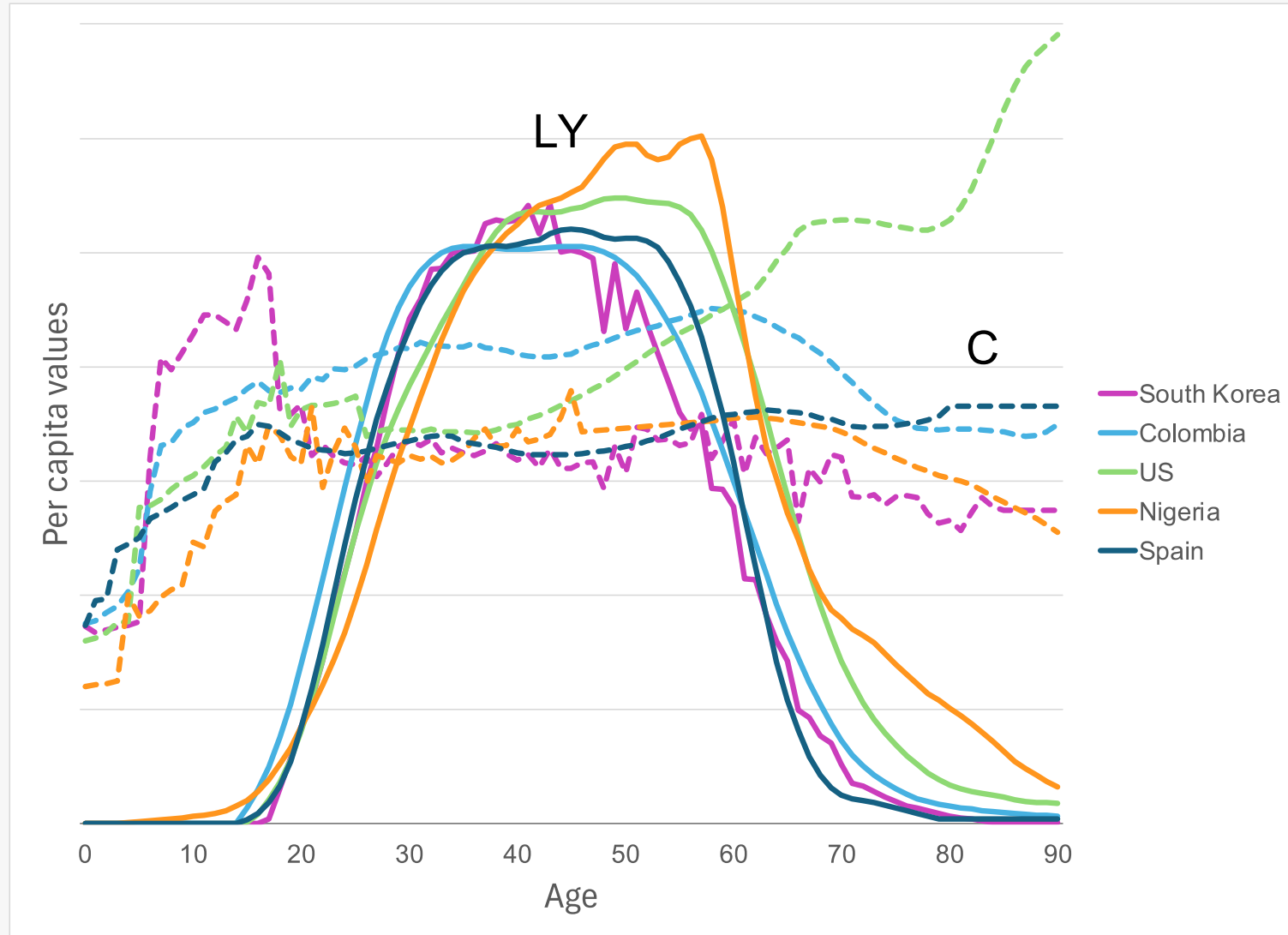
National Transfers Accounts project



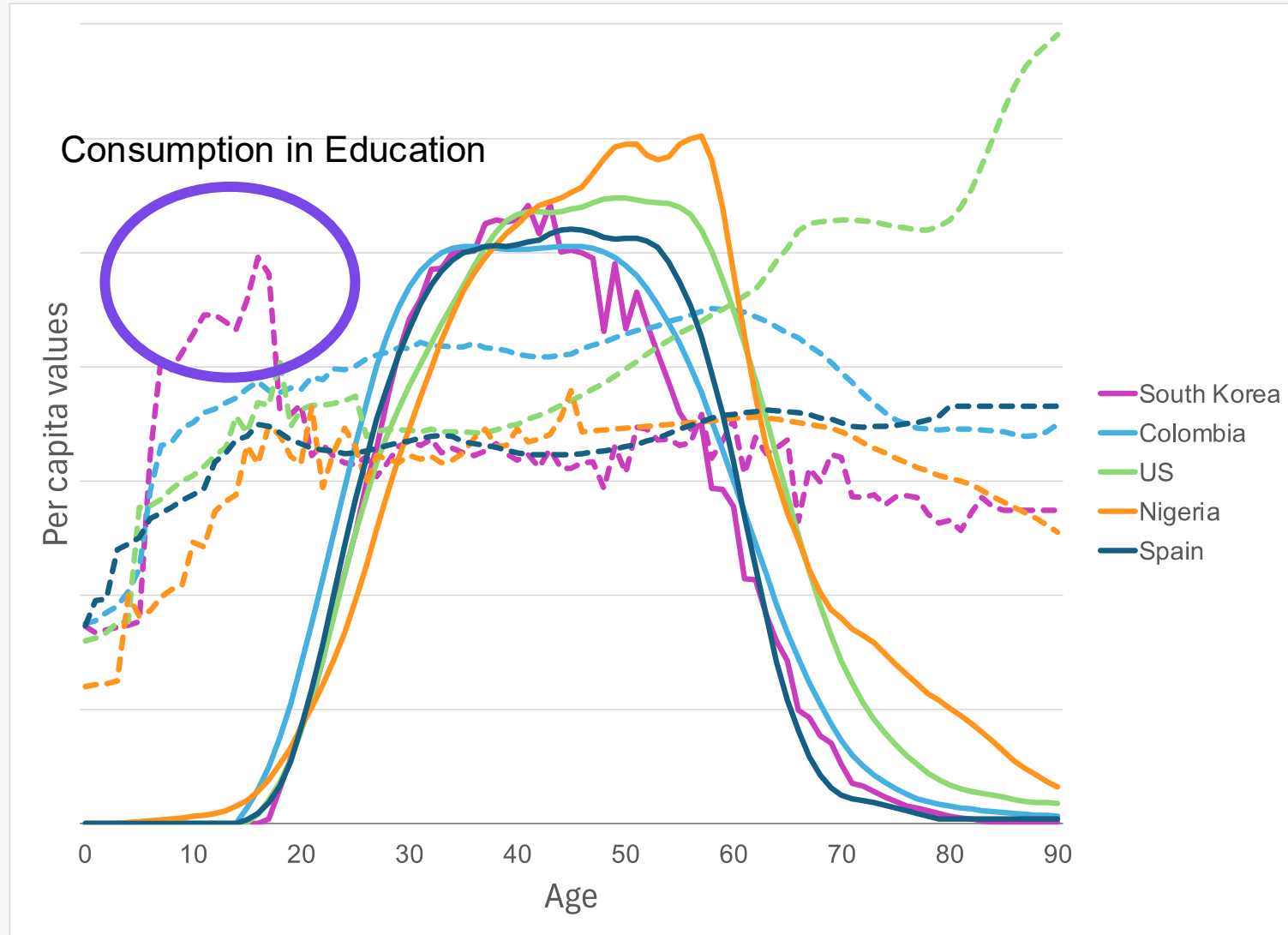
- www.ntaccounts.org
- More than 100 countries participating
- National teams
- Several international projects
- European data available at: AGENTA
- United Nations in charge of publishing the methodological manuals



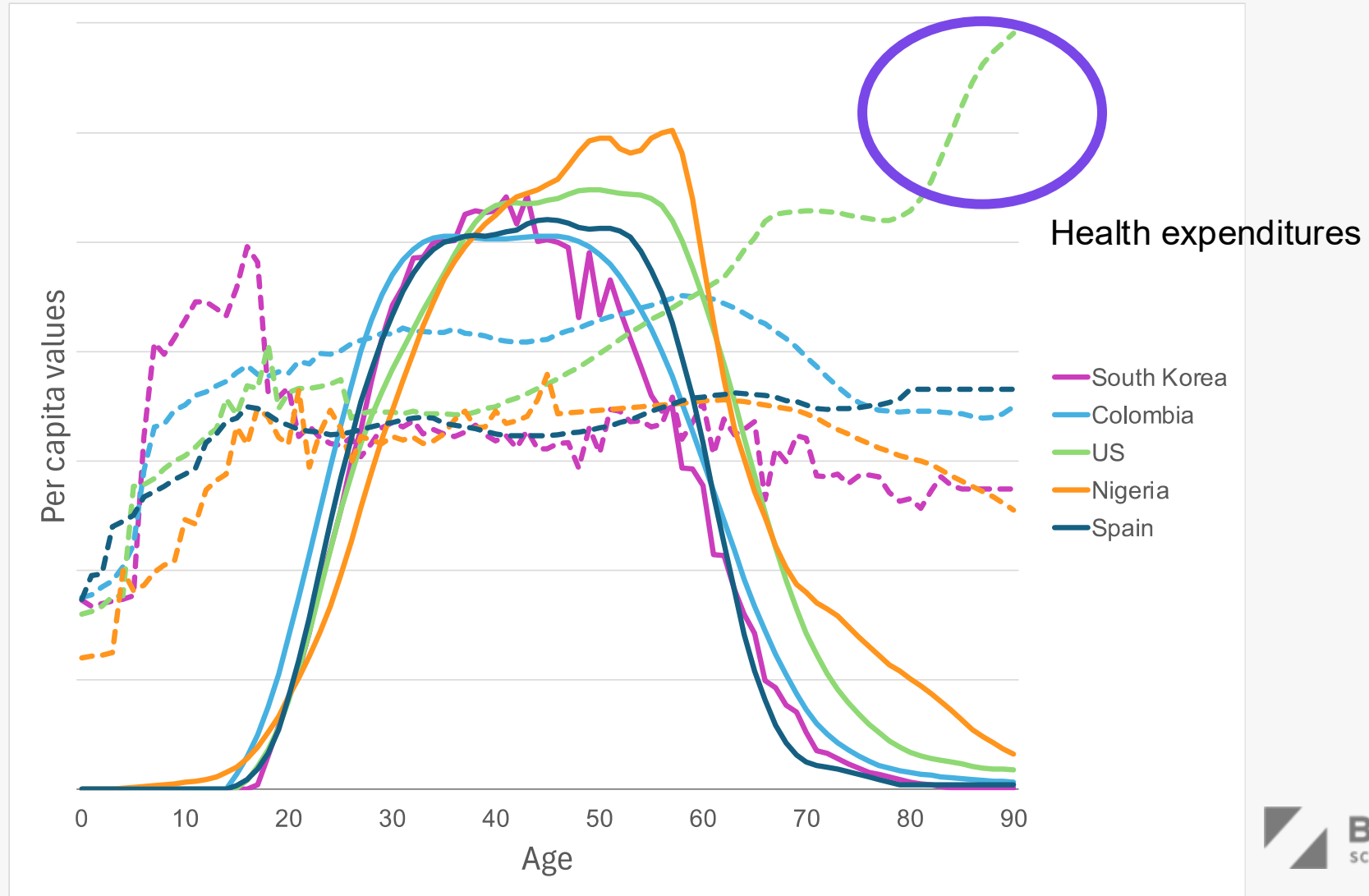
Consumption and Production profiles across countries



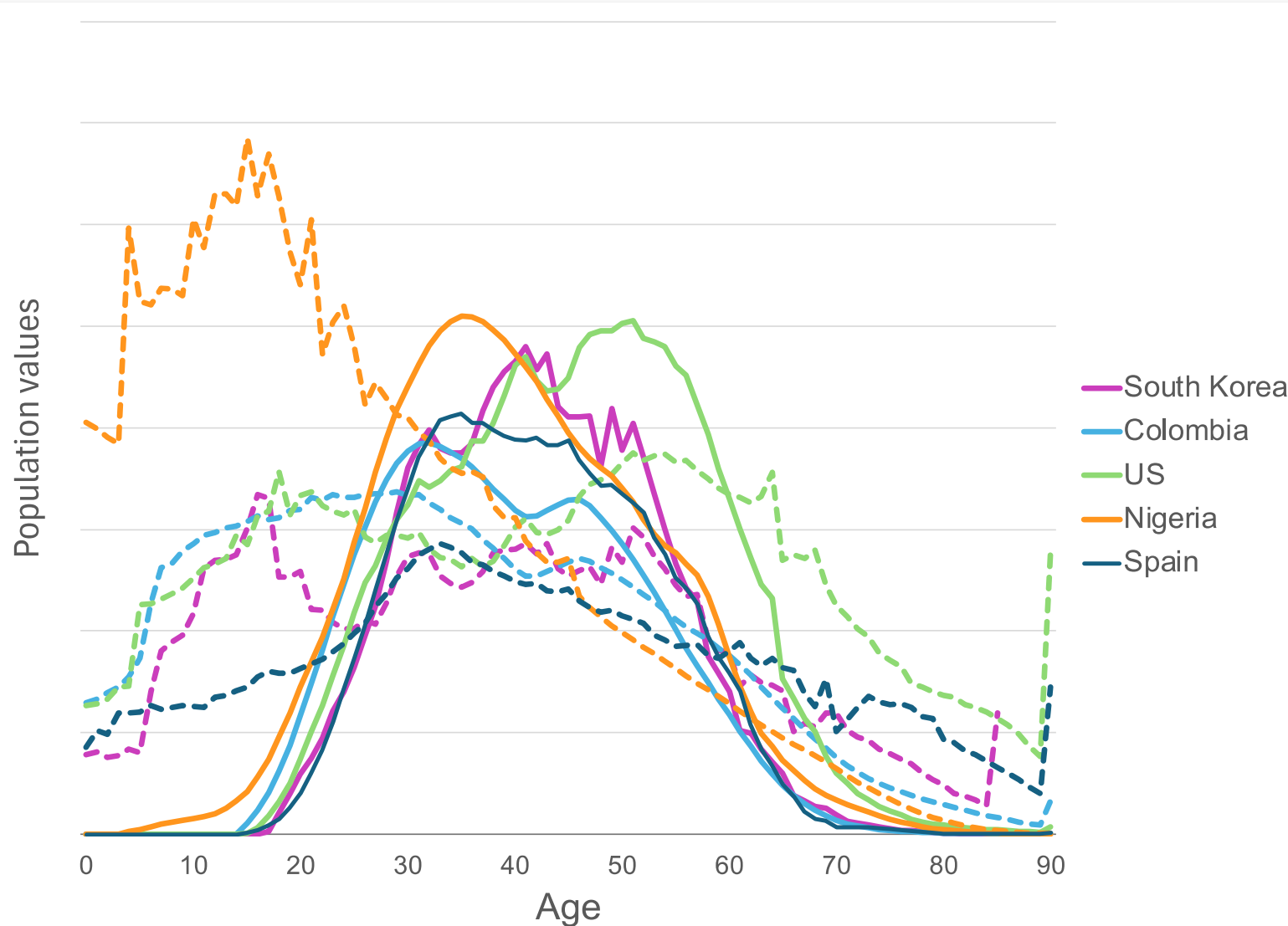
Consumption and Production profiles across countries



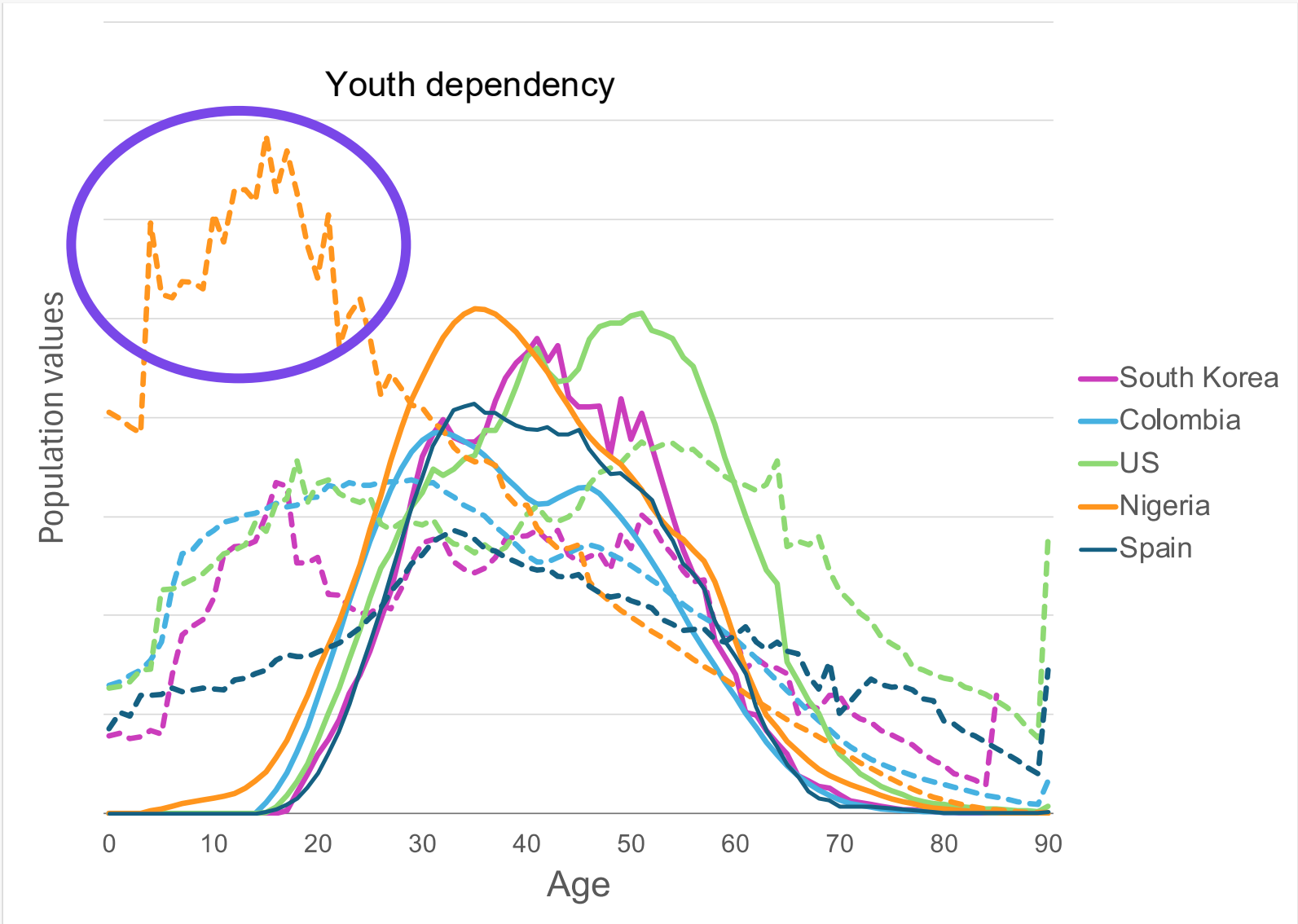
Consumption and Production profiles across countries



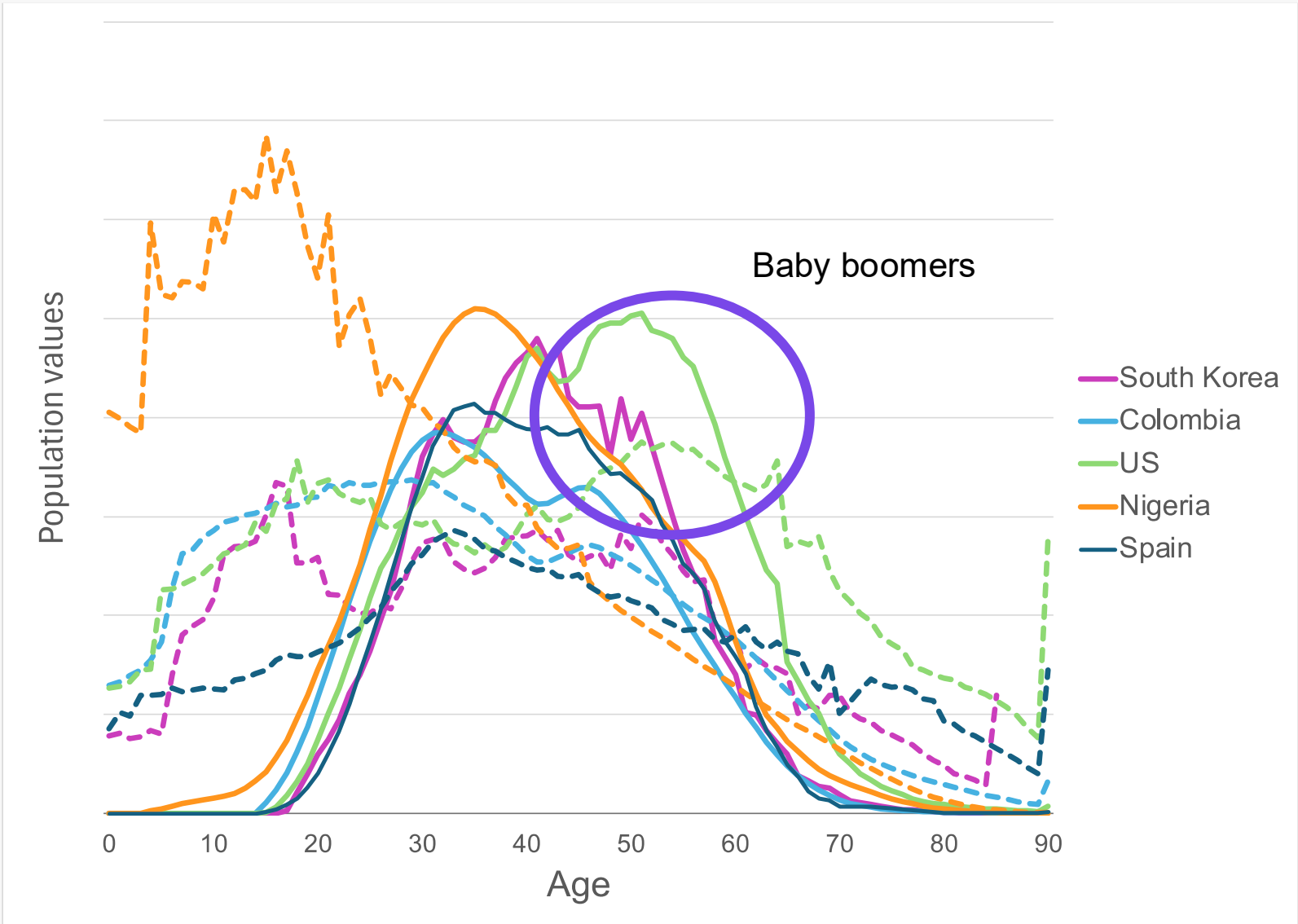
The importance of the age structure



The importance of the age structure



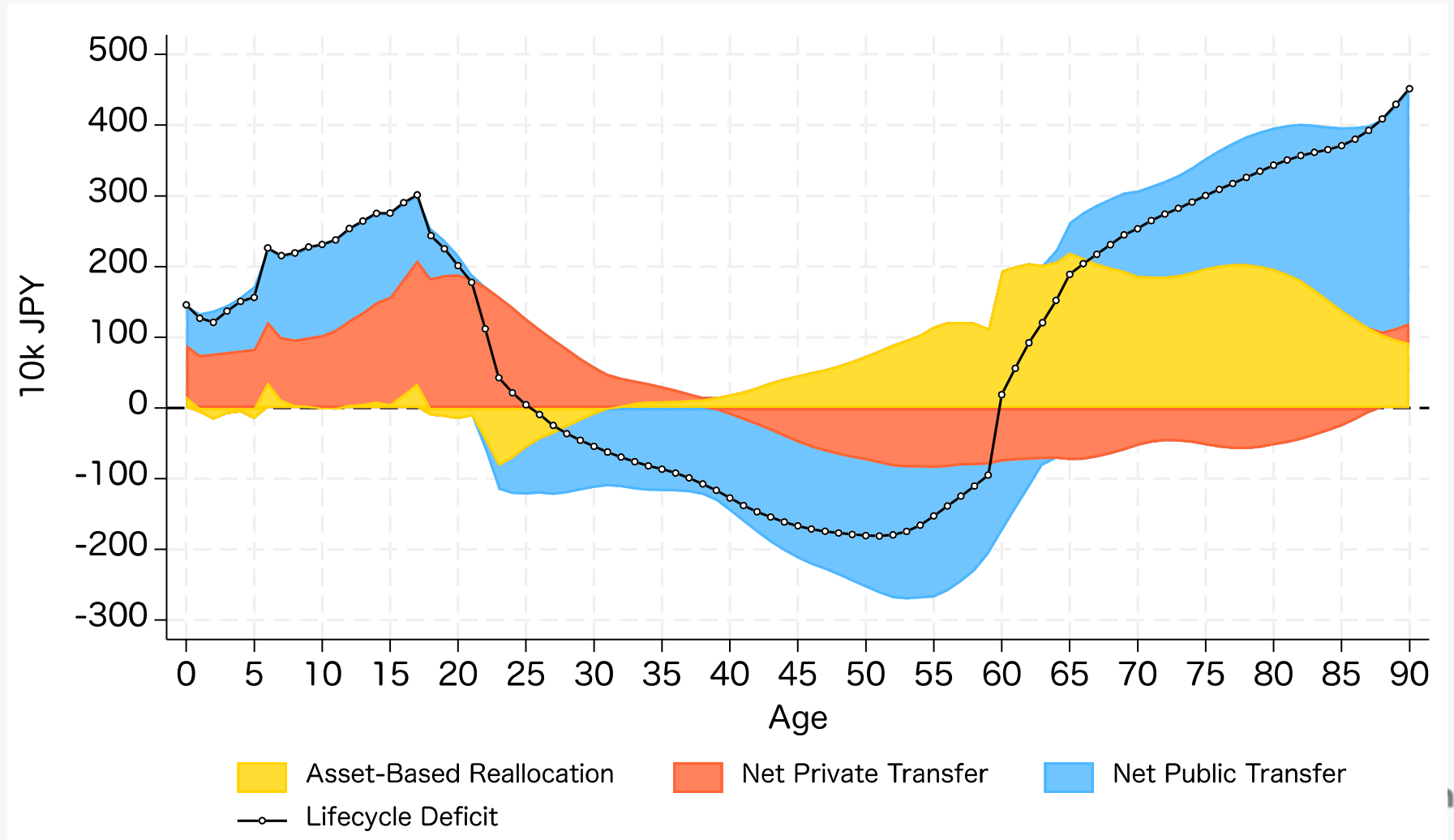
The importance of the age structure



Financing the Lifecycle Deficit by age

$$LCD = C - LY$$

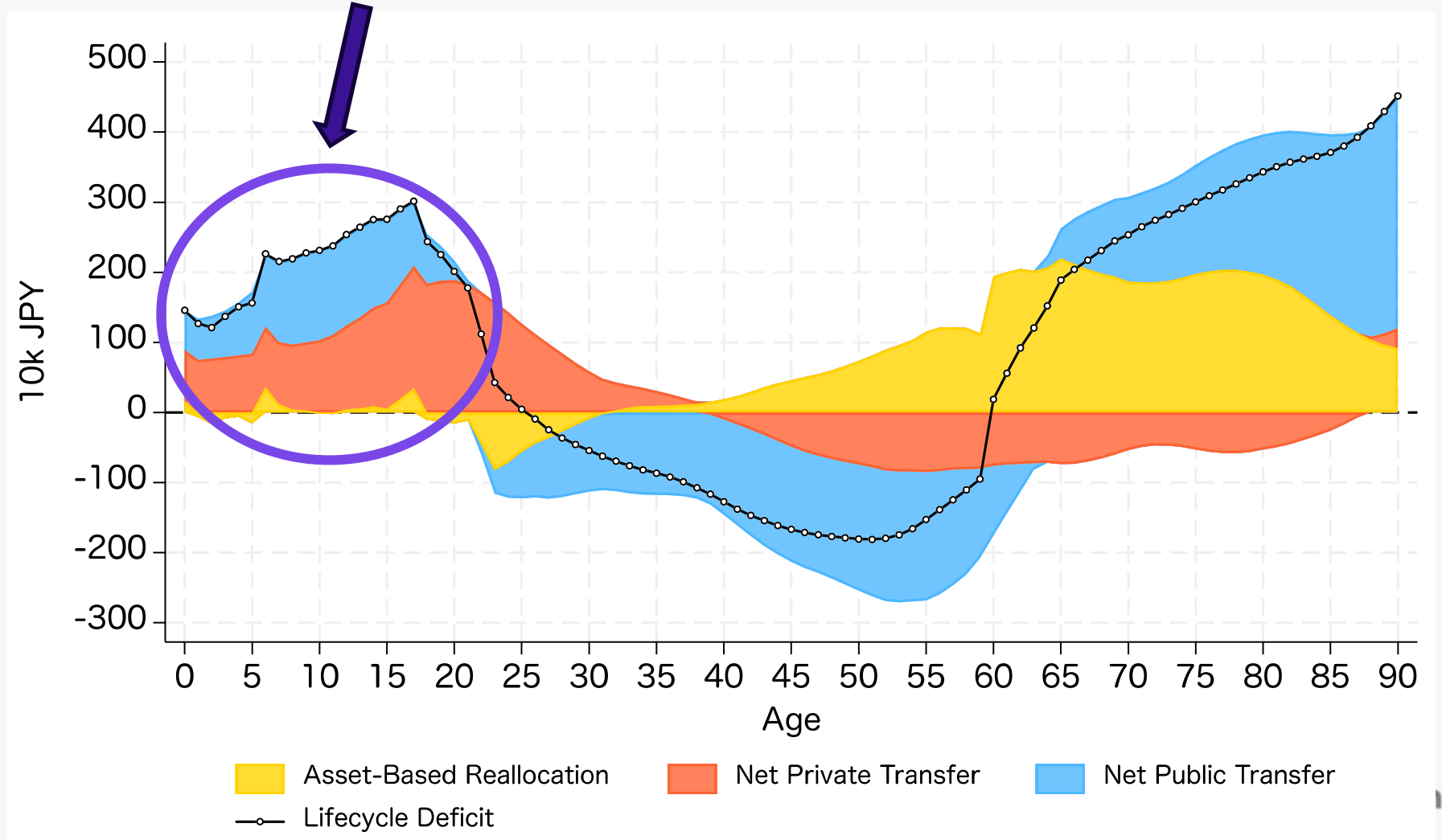
JAPAN 2014



Financing the Lifecycle Deficit by age

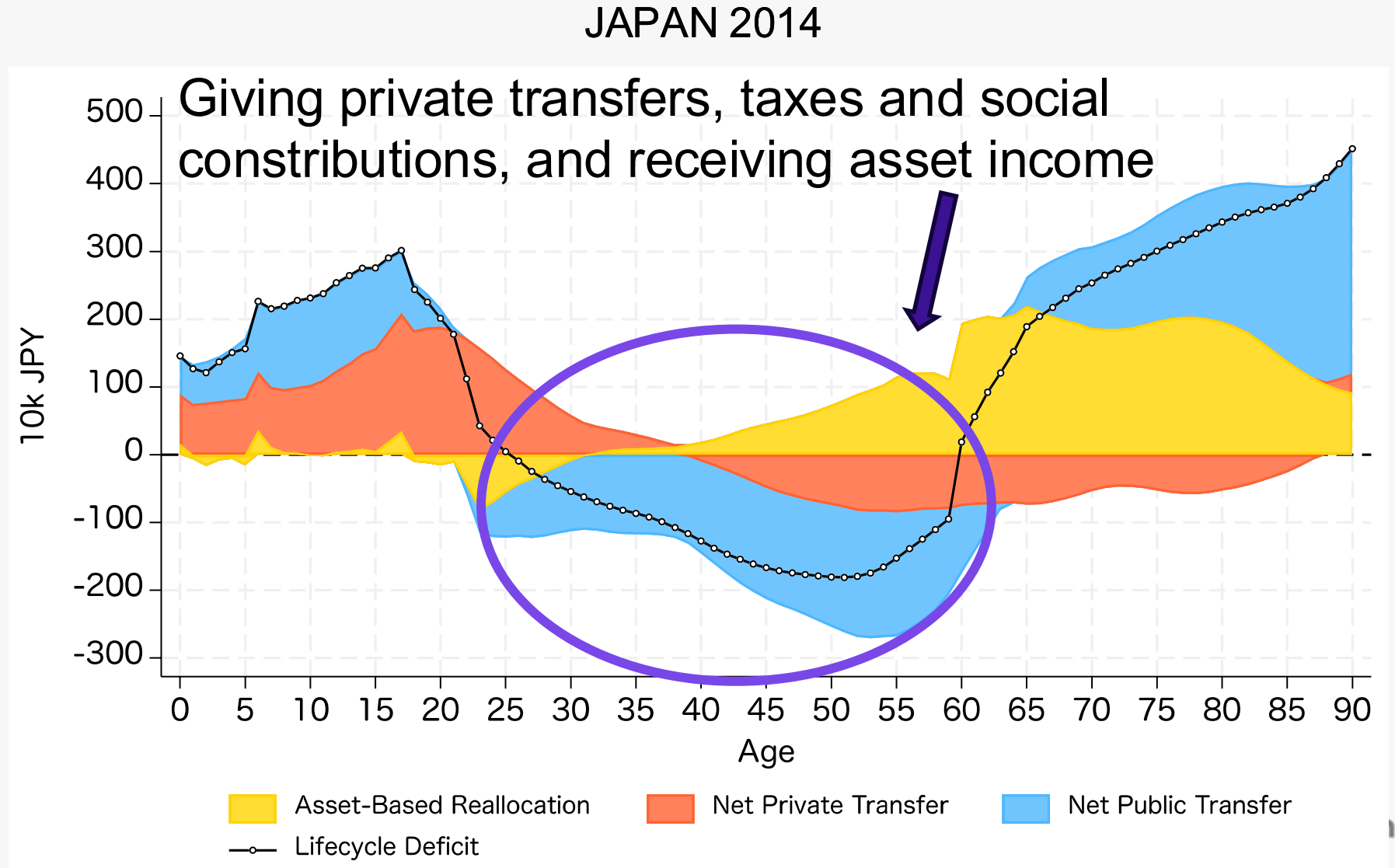
$$LCD = C - LY$$

Receiving Education, Health and General Consumption JAPAN 2014



Financing the Lifecycle Deficit by age

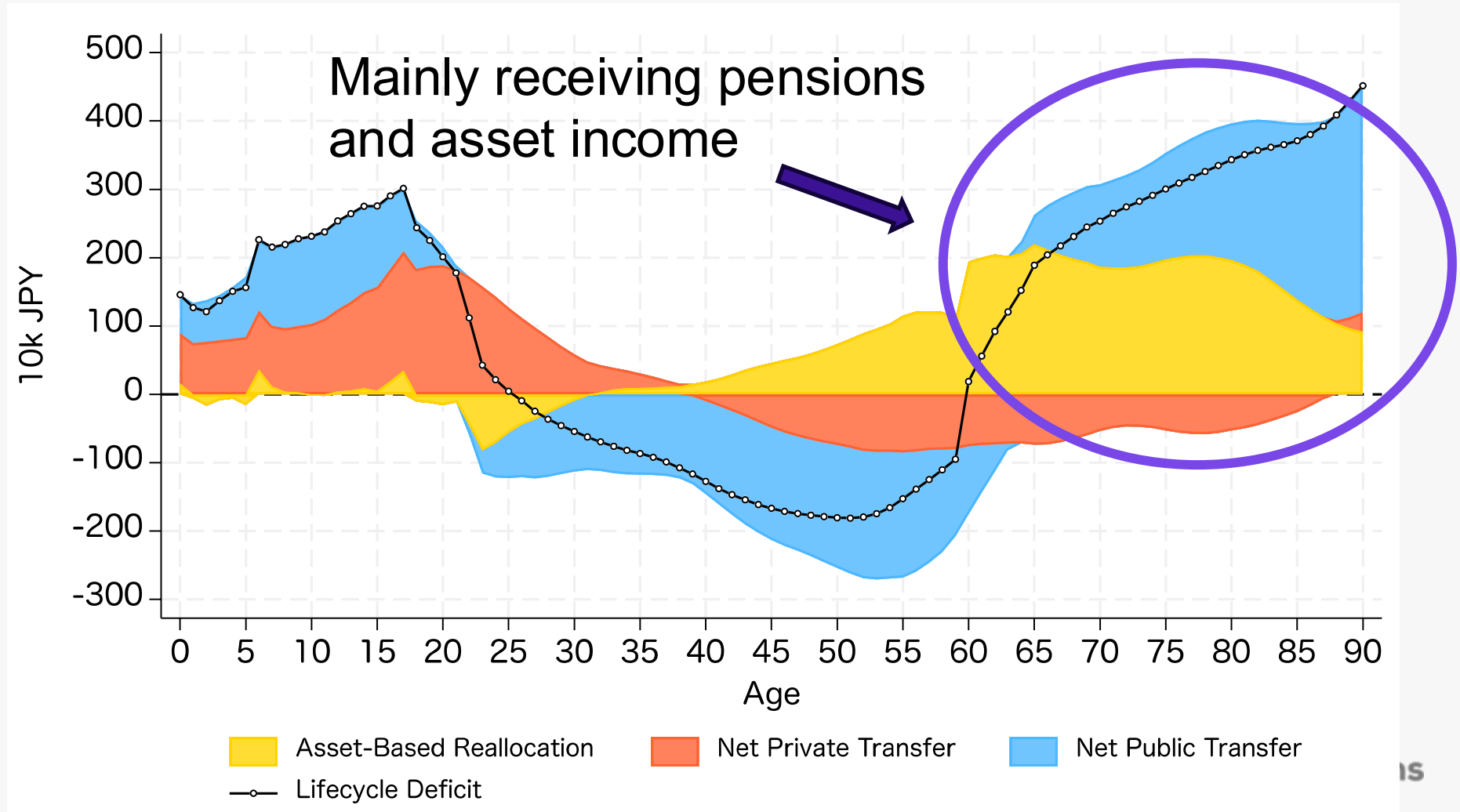
$$LCD = C - LY$$



Financing the Lifecycle Deficit by age

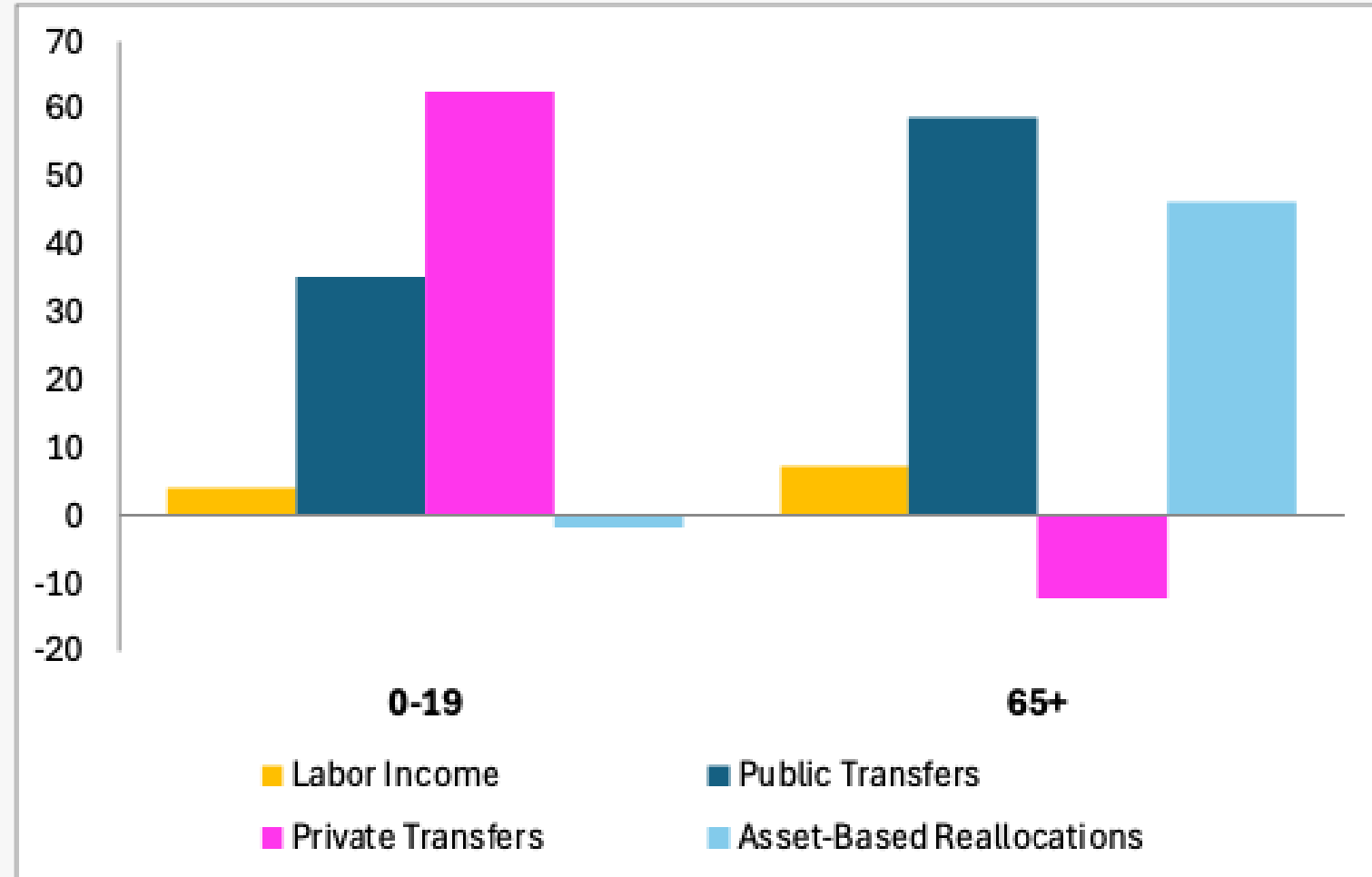
$$LCD = C - LY$$

JAPAN 2014

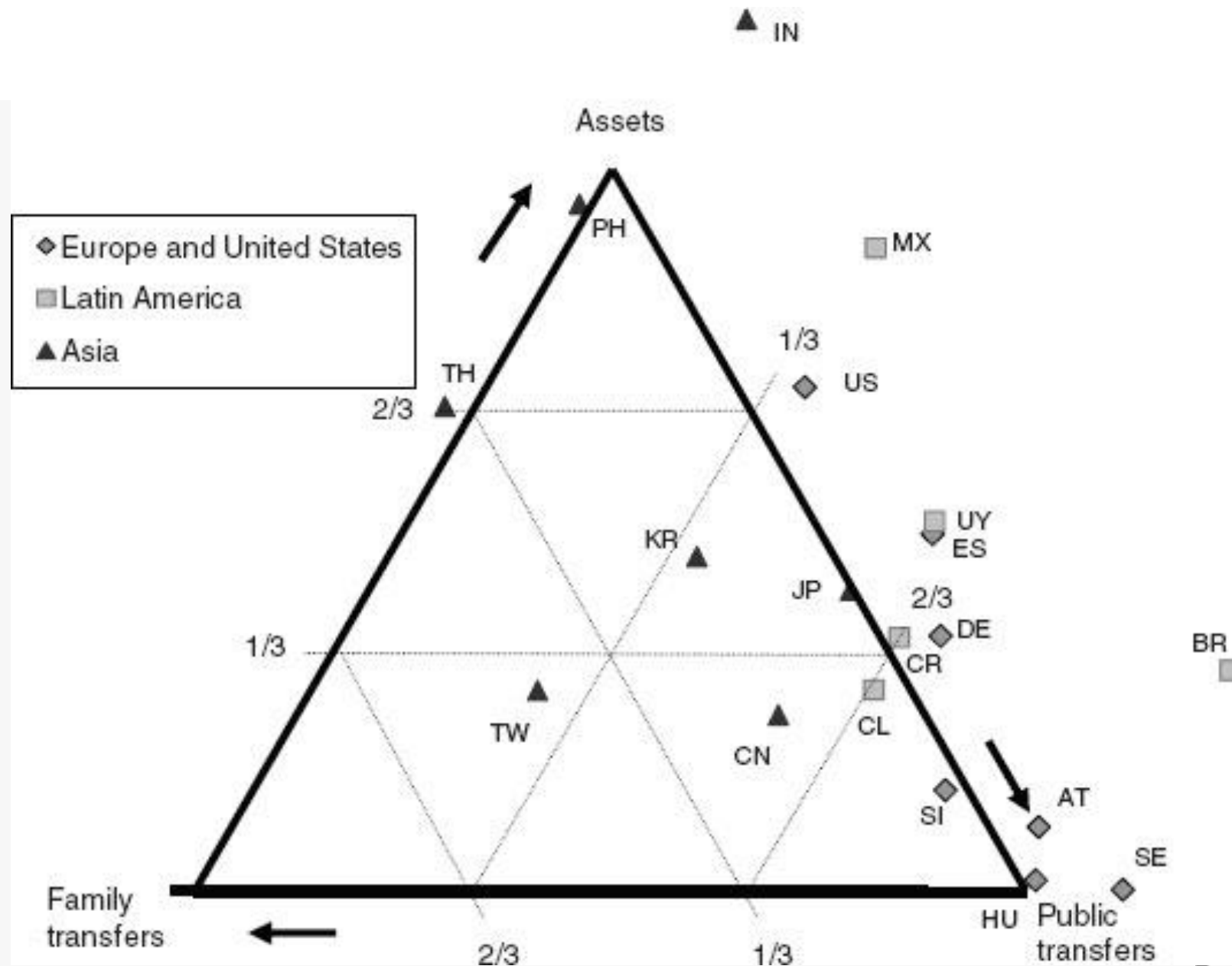


Financing the Lifecycle Deficit in “dependent” age groups

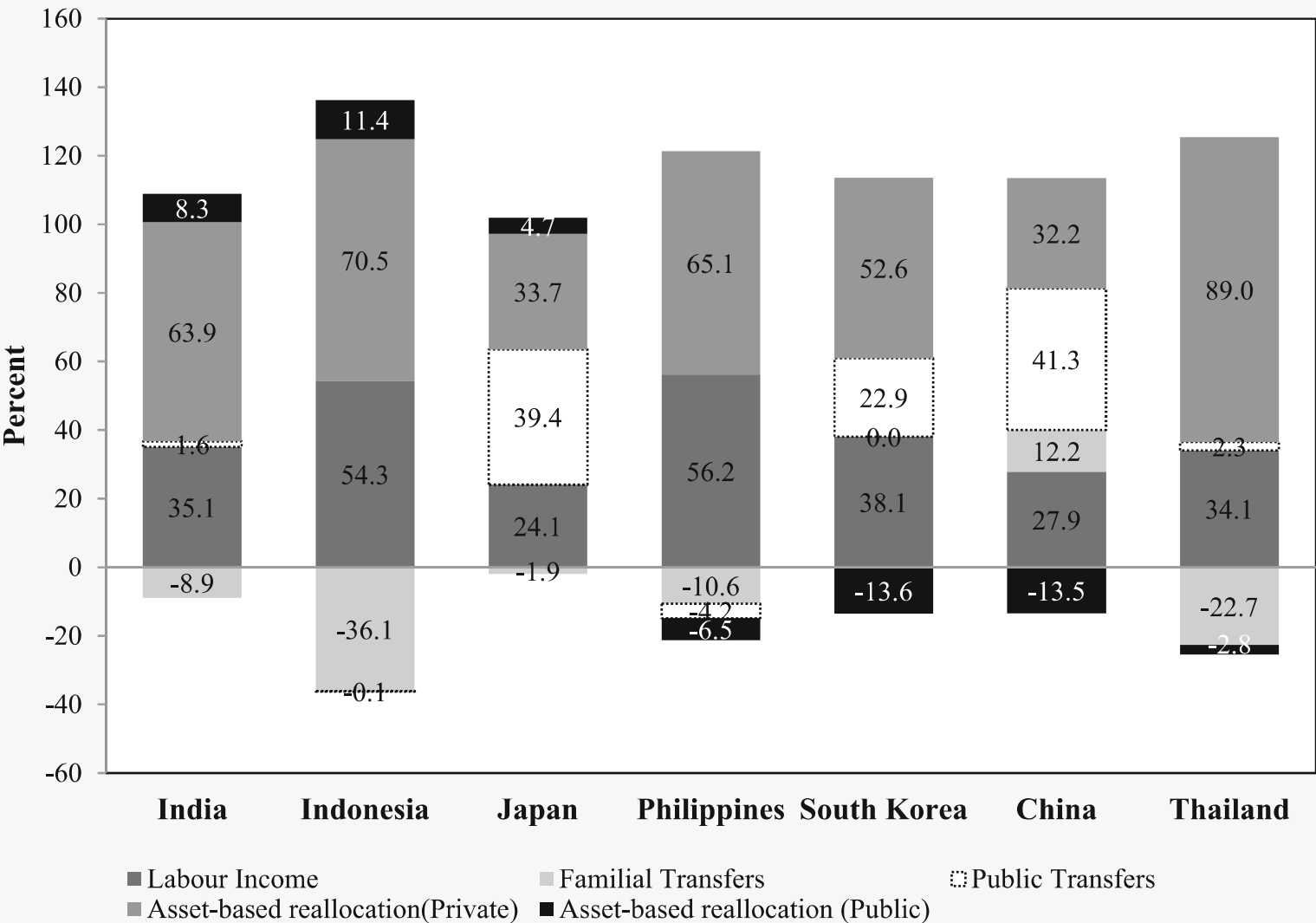
SPAIN 2000



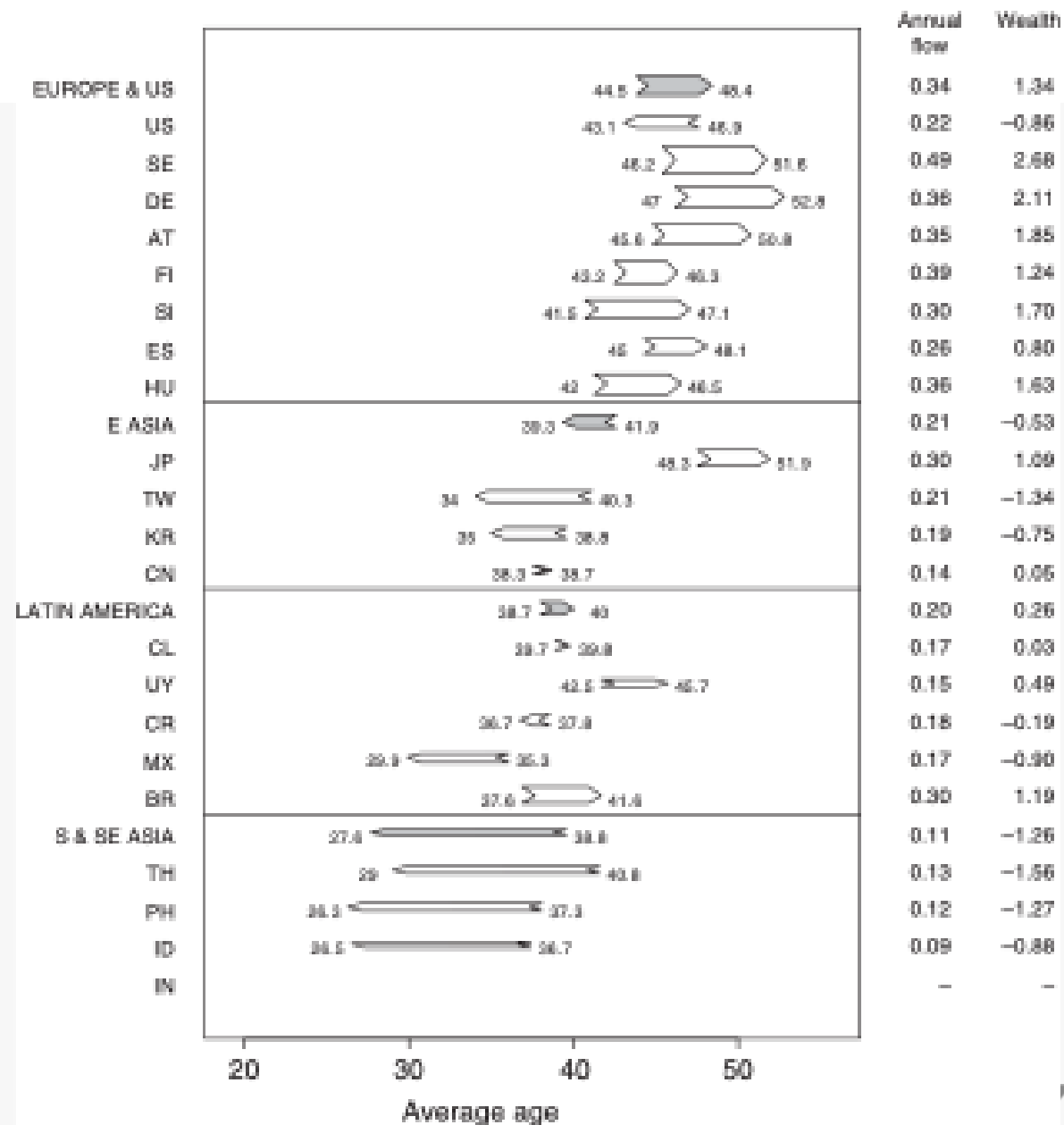
Triangle



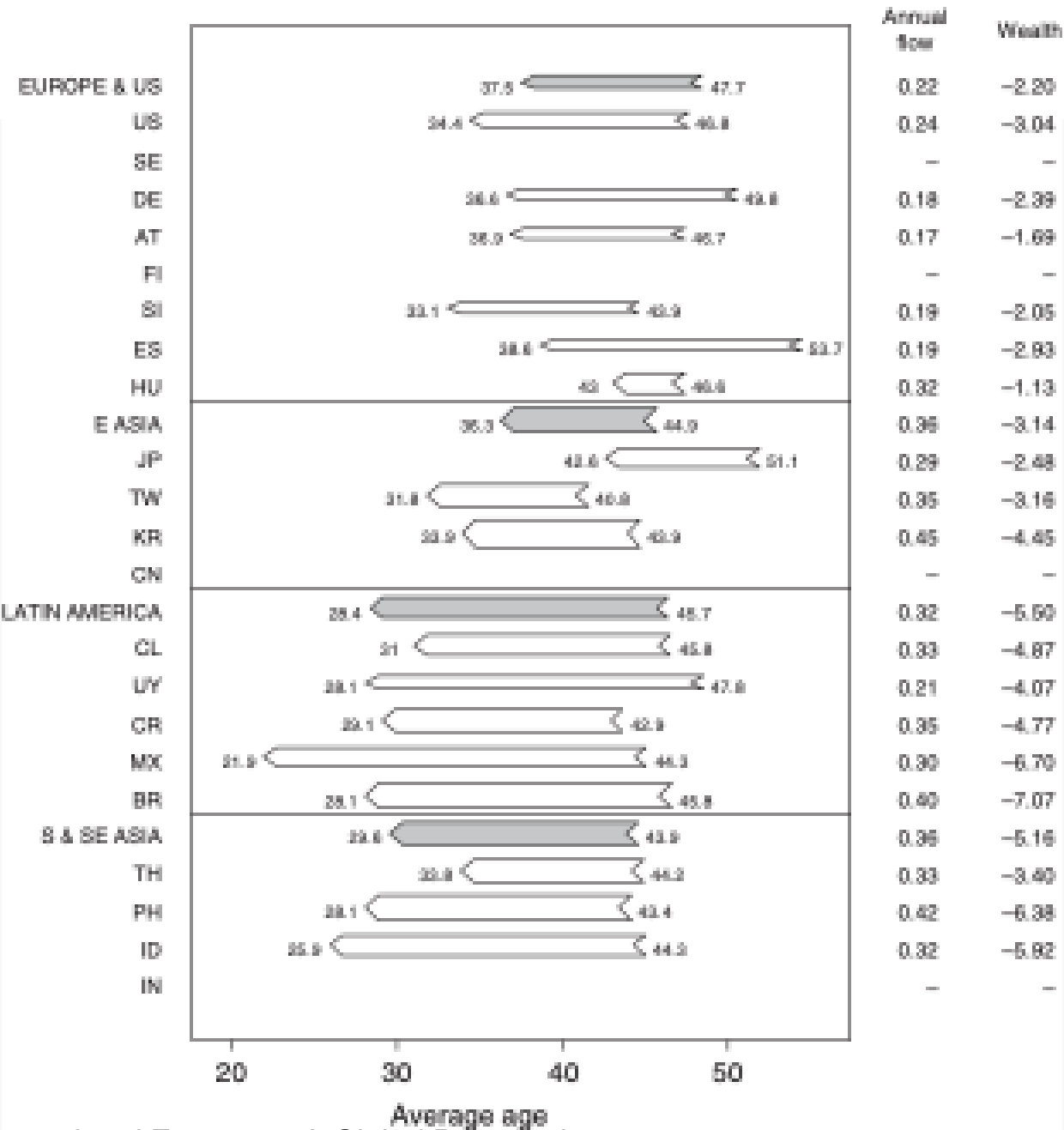
Financing the Lifecycle Deficit of older people in Asia



Wealth Arrows of Public Transfers



Wealth Arrows of Private Transfers



Aspects that Affect the Economic Sustainability of Population Aging



Pension systems

Pay-as-you-go systems (PAYG)

- Financed by the current working population
- No accumulation of wealth

Funded systems (Capitalization)

- Individual funding through contributions of workers
- Public/Private
- Employees/Employers

Non-contributory pensions

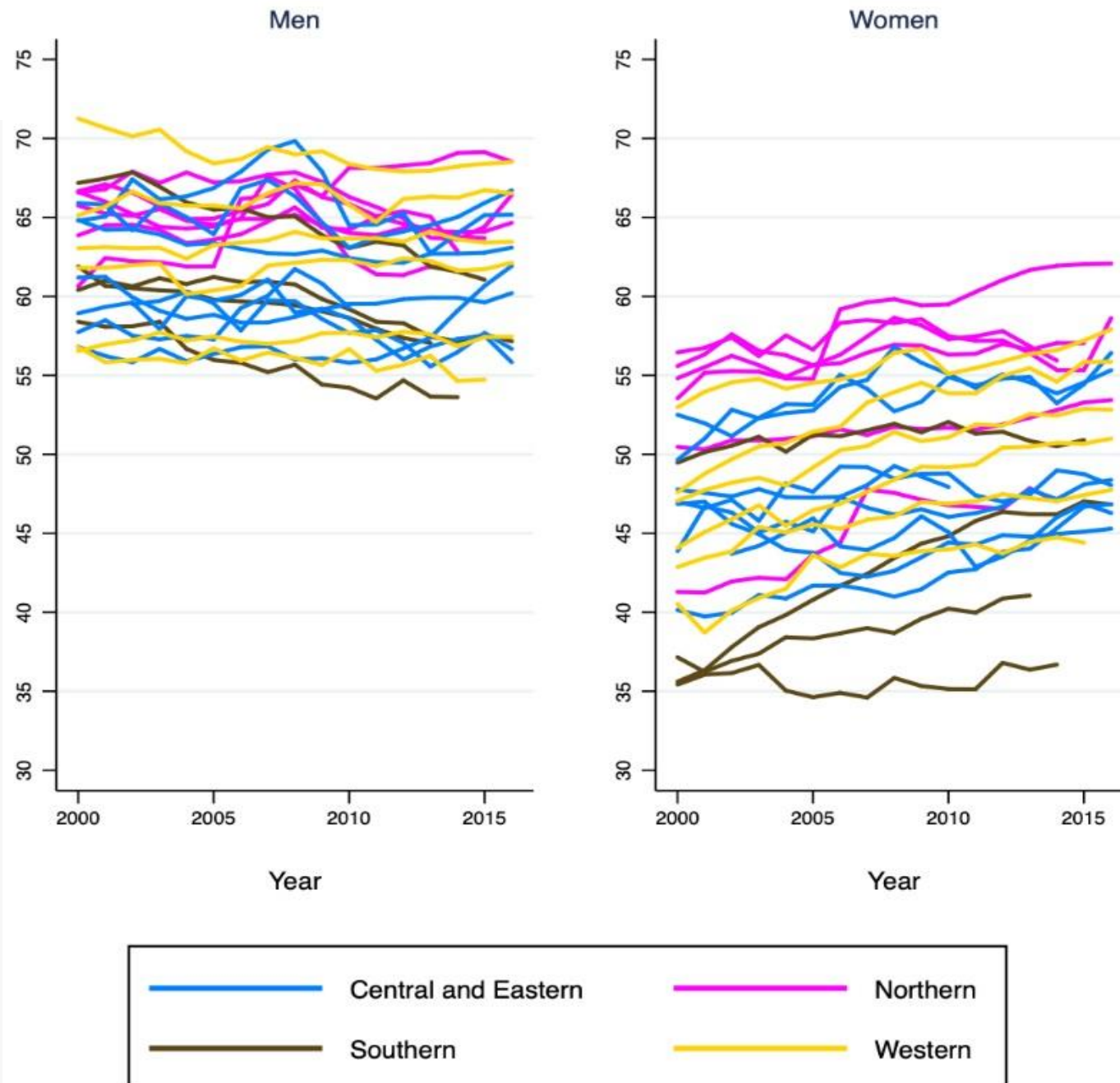


Labor Activity

- Gender gaps are still large



Working Life as a proportion of Life Expectancy in Europe



Labor Activity

- Gender gaps are still large
- Differences in hours worked – impact in income levels



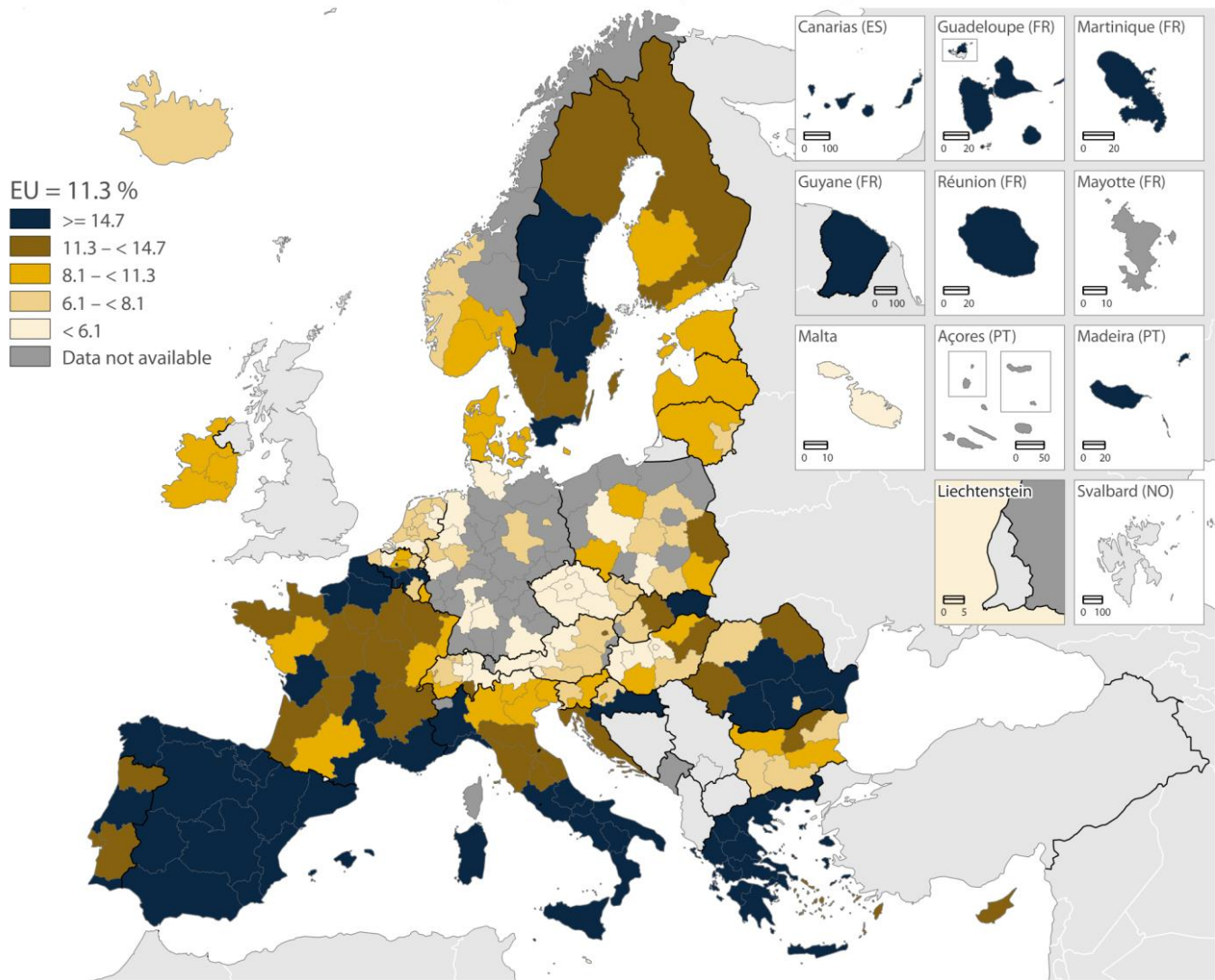
Labor Activity

- Gender gaps are still large
- Differences in hours worked and impact on income levels
- Youth unemployment rates



Labor Activity

Youth unemployment rate (15-29 years), 2022
(in % of the labour force, people aged 15-29, by NUTS2 region)



Eurostat online data code lfst_r_lfu3rt

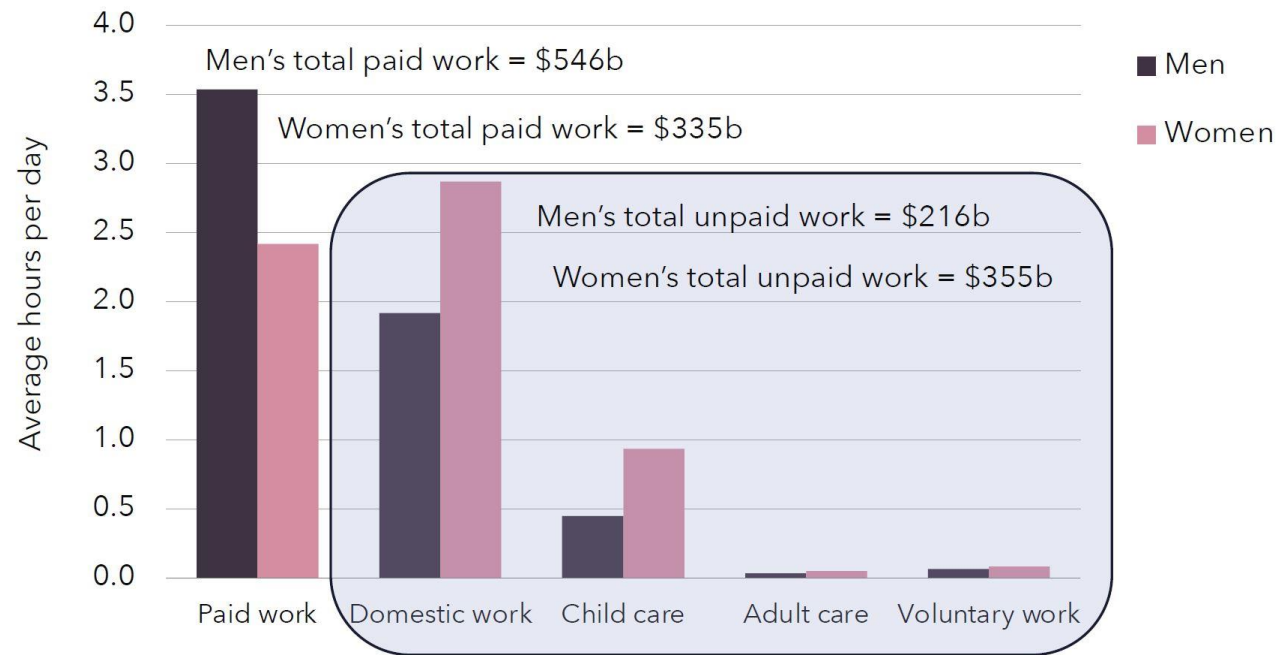
Administrative boundaries: © EuroGeographics © UN-FAO © Turkstat
Cartography: Eurostat – IMAGE, 05/2023



Unpaid production

- Care and Domestic work
- What if we treat them as productive activities?

Equivalent total annual labour value of paid and unpaid productive activities



Source: ABS Time Use Survey, 2021. Average hours spent on activity include people who do not participate in the activity at all. Equivalent labour value expressed as billions of dollars and aggregated for the population annually.



Labor productivity

- Level of income per hour worked
- Human capital: education and health
- Manual/Non-manual jobs
- Precarious work
- What about AI?



Asset income and Wealth

- Increasing Wealth inequalities
- Higher concentration of wealth at older ages
(housing wealth)



Alternative scenarios for demographic sustainability



Gender

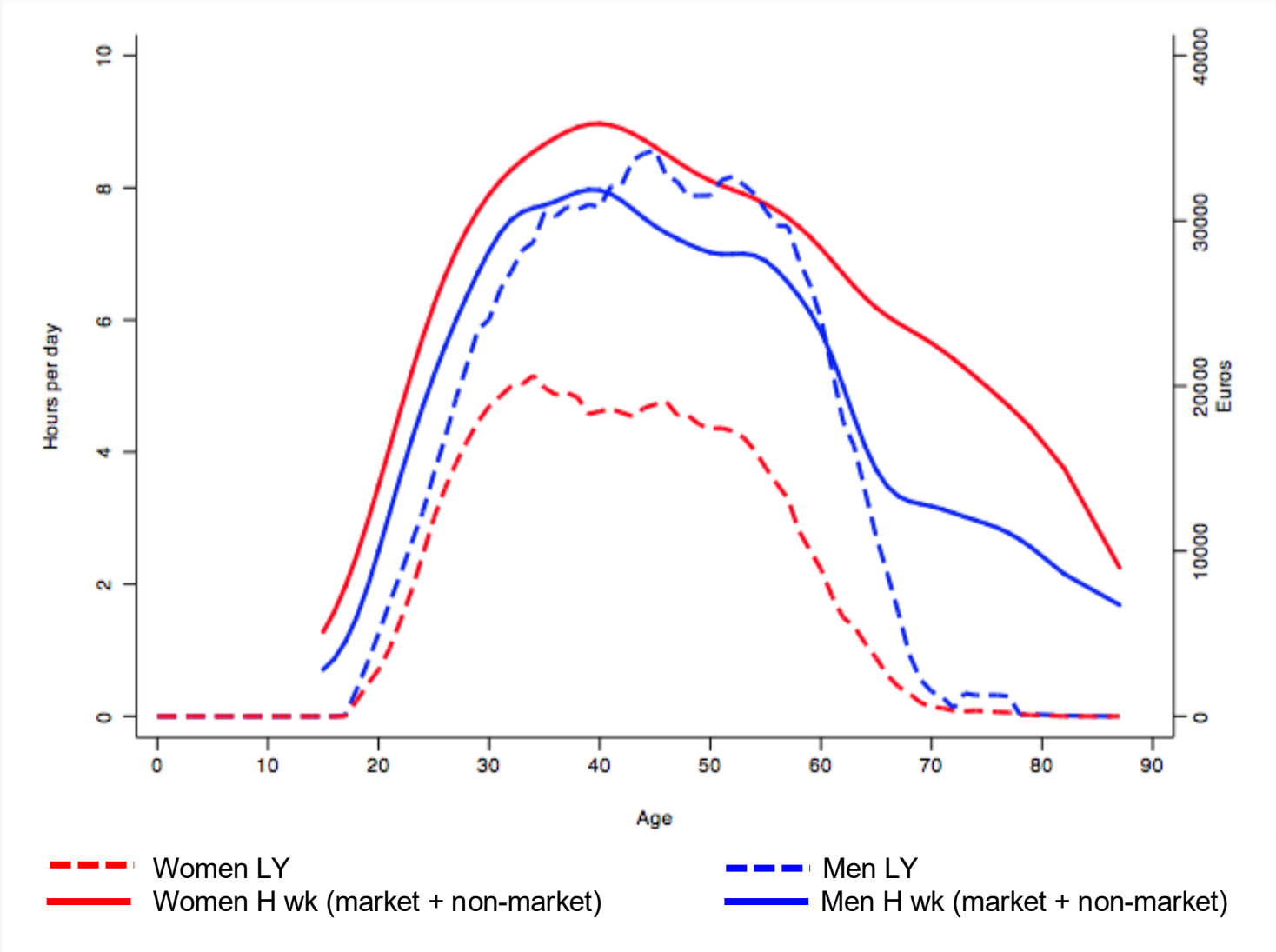
What if we take into account two things:

- Recognize and pay for unpaid production work
- If we assume an increase in labor participation and labor income of women



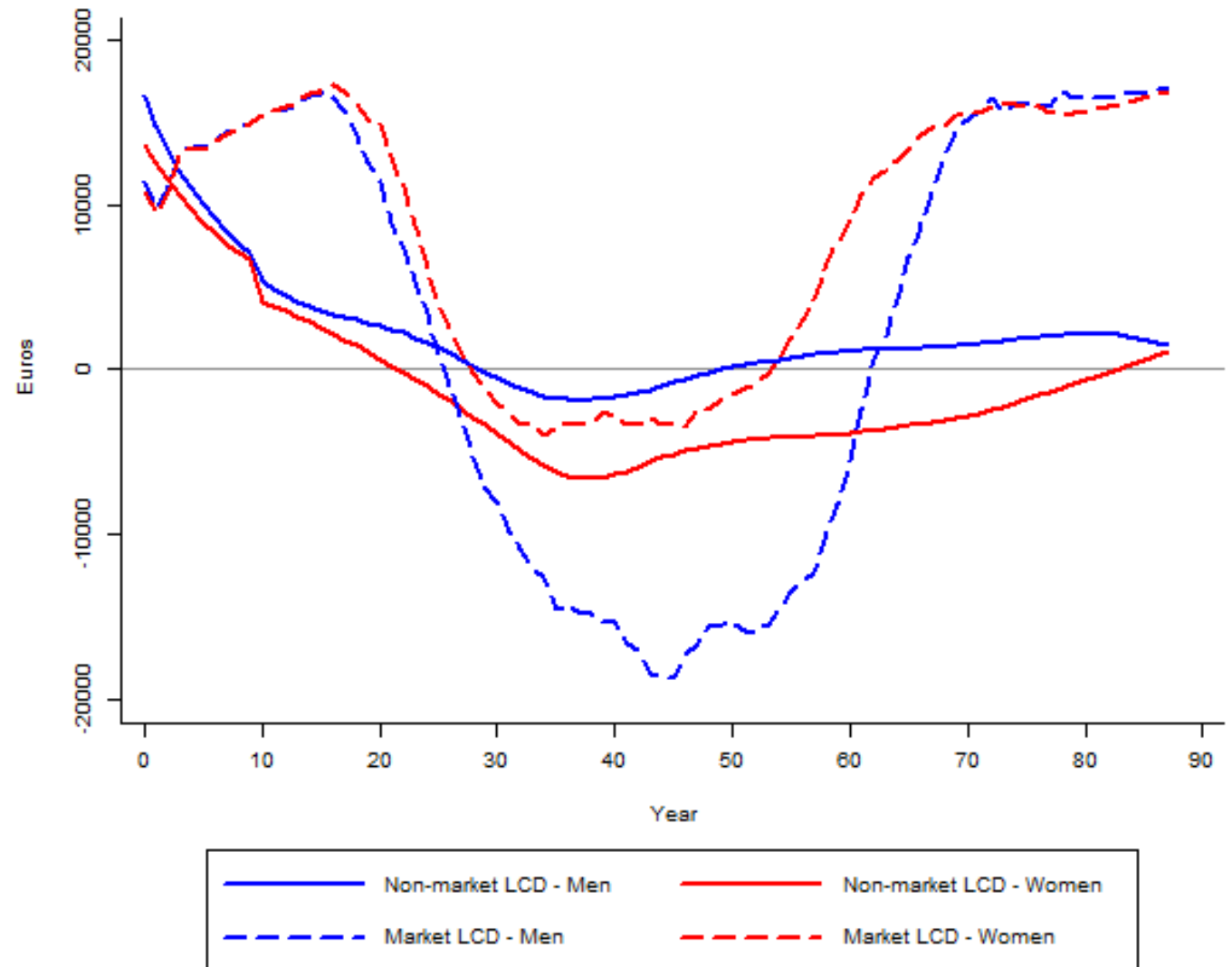
Gender

SPAIN 2009



Rentería et al., 2016

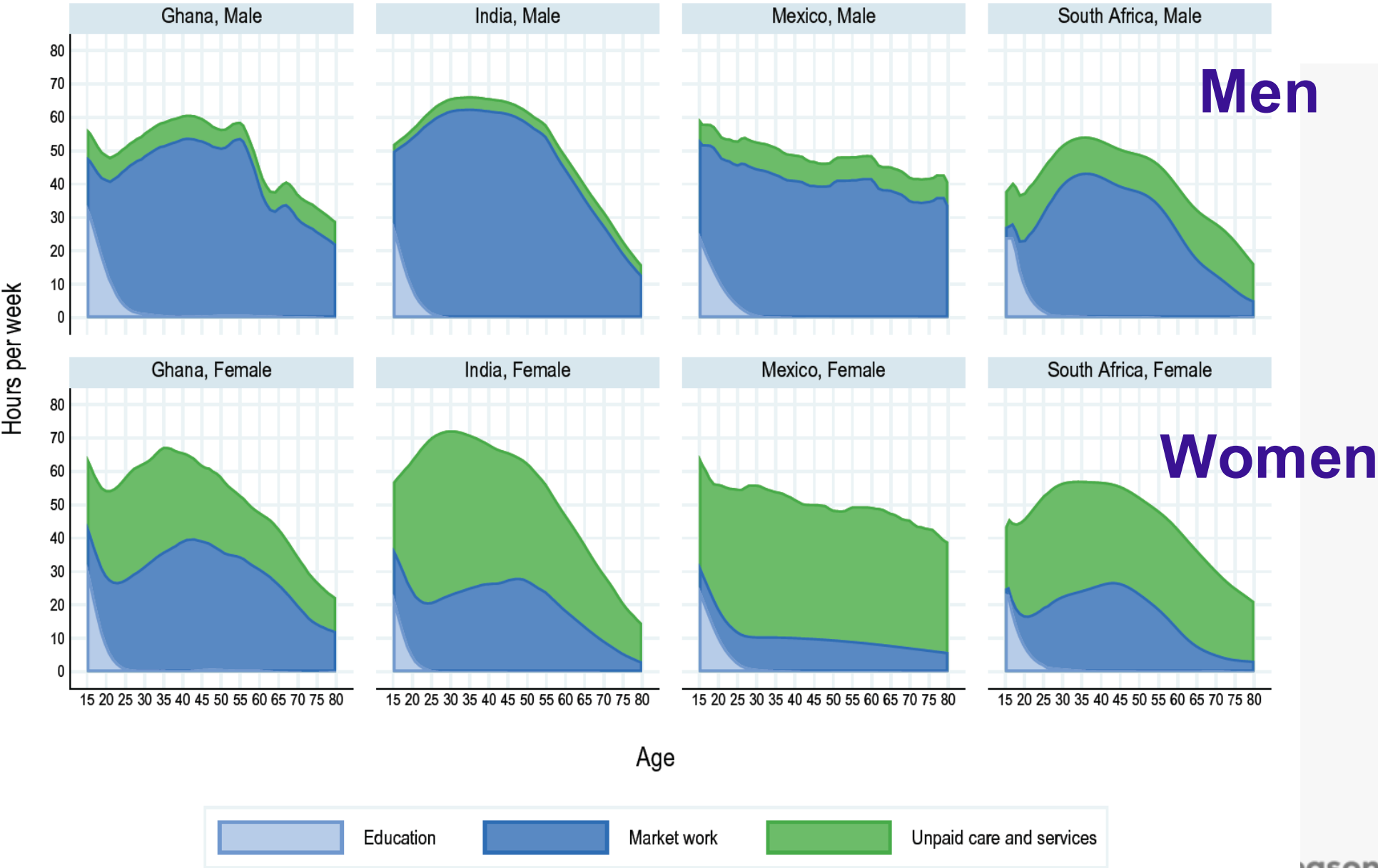
LifeCycle Deficit Market and Non- Market



Source: Author's own calculations



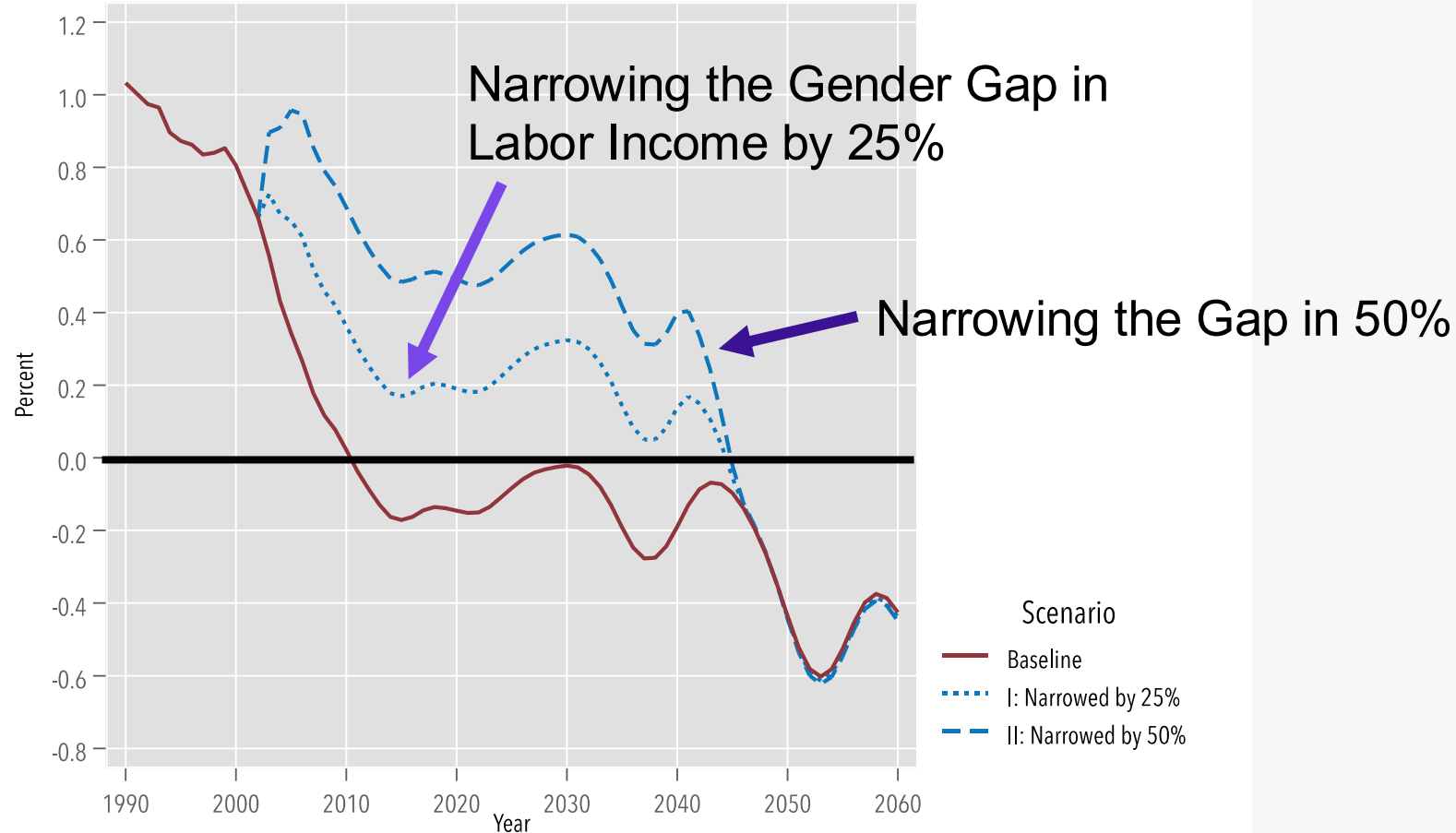
Counting Women's Work Project



Source: Calculated from data available on the NTA Counting Women's Work project website at <http://www.countingwomenswork.org/>.



Demographic Dividend in Mauritius



Source: Reproduced from Oosthuizen (2018). Notes: Estimates assume a narrowing of the gap between the male and female labour income profiles of 25 percent or 50 percent over a 40-year period from 2003 to 2043.

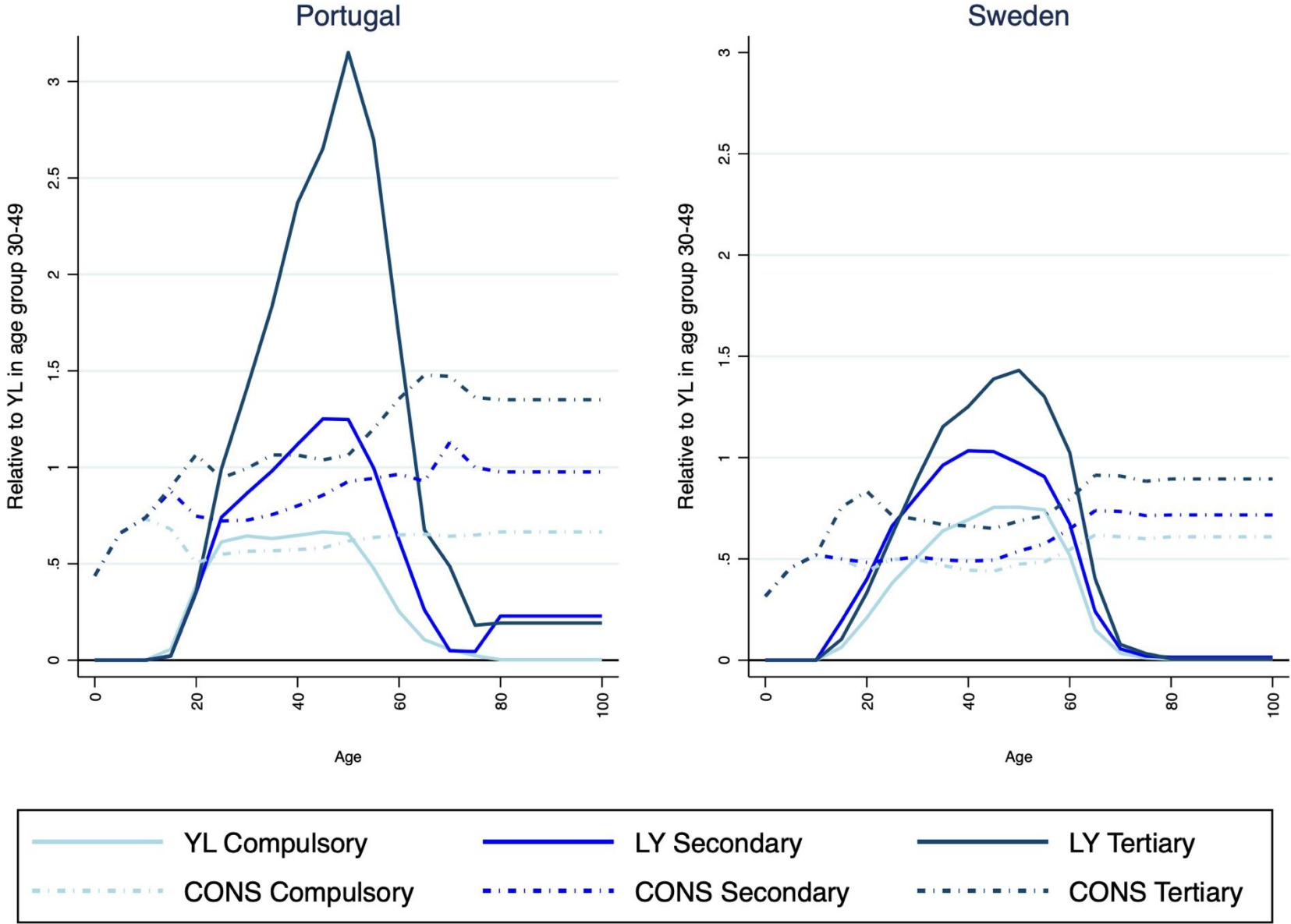


Education

- Economic profiles differ by socioeconomic status (as educational levels)
- For example, education expansion can have a positive impact on the economic sustainability of the support ratio

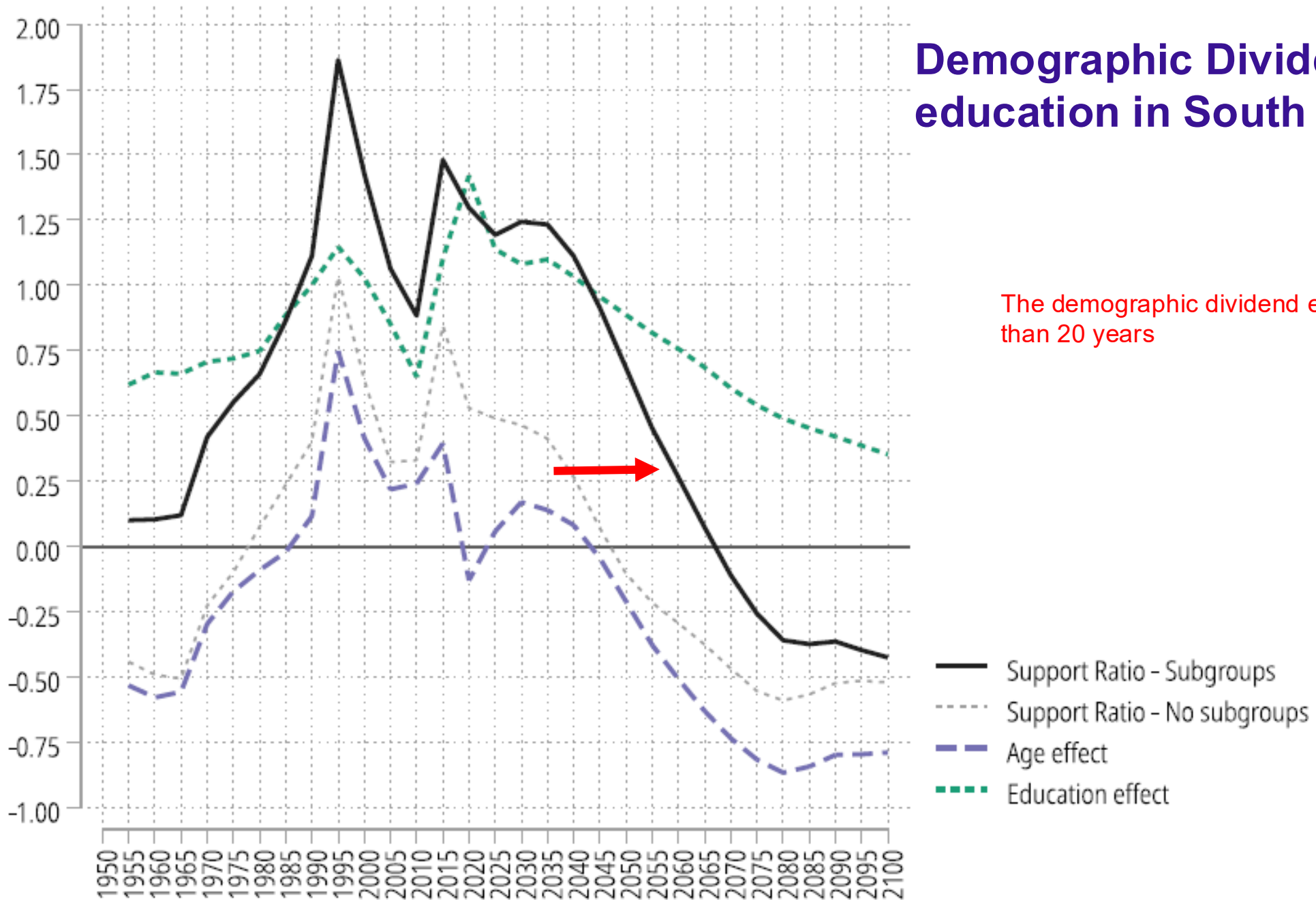


Education inequalities in economic profiles



Demographic Dividend by education in South Africa

Average annual growth rate



The demographic dividend expands more than 20 years



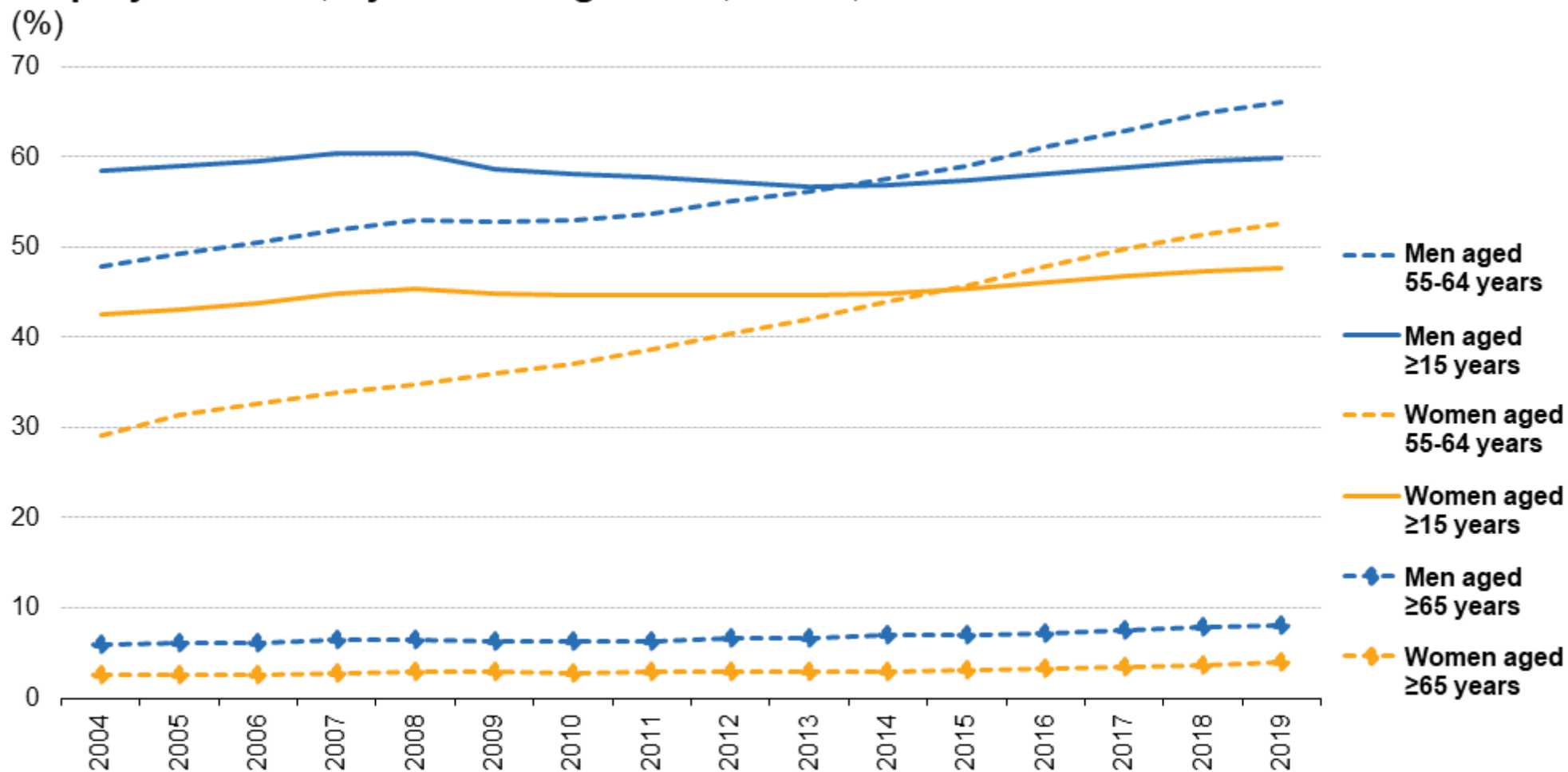
Health

- Improvements in health and activity of older individuals
- Called the **Silver dividend**
- The reality is that older individuals do not want to extend their labor
- It depends greatly on the extension of the retirement age
- This cannot apply to all jobs



Health

Employment rate, by sex and age class, EU-27, 2004-2019



Note: the indicator is defined as the number of people of a certain age who are in employment divided by the total population of the same age group. Breaks in series: 2005.

Source: Eurostat (online data code: lfsa_ergan)



Exercise



Calculate how to finance the LifeCycle Deficit (LCD) by education level

- You have the age profiles of Consumption, Labor Income, Public Transfers, Private Transfers and Asset-Based Reallocations
- Discover the age limits of the Lifecycle deficit (C-LY), of Public Transfers, Private Transfers and Asset-Based Reallocations.
- How children (0-18) finance their LCD?
- How older adults (65+) finance their LCD?



Financing the Lifecycle Deficit (LCD) or Consumption

$$C - YL = (TG^+ - TG^-) + (TF^+ - TF^-) + (YA - S)$$

- These profiles should equal at each age, but also at each age group.
- Calculate the % of the LCD that corresponds to Net Public Transfers, Net Private Transfers and Asset-Based Reallocations.
- By leaving the Consumption alone in one side of the equation, you can also calculate the % to finance Consumption.



Gràcies

Elisenda Rentería
erenteria@ced.uab.cat

Júlia Almeida Calazans
jalmeida@ced.uab.cat

Mariana Calderón-Jaramillo
mcalderon@ced.uab.cat

